

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY(DU)
School of Computer Engineering and Mathematical Sciences
M.Tech (Data Science) End Semester- Autumn 2022 Examination

Entry No: - -
 Date: 27-12-2022

Total Number of Pages: [02]
 Total Number of Questions: [05]

Course Title: Data Structures and Algorithms with C
 Course Code: AM 607D

Time Allowed: 3.0 Hours

Max Marks: [50]

Instructions / NOTE

- i. Attempt All Questions.
- ii. Support your answer with neat freehand sketches/diagrams, wherever appropriate.
- iii. Assume any missing data to suit the case / derivation / answer.
- iv. Following calculators or lower models are allowed:
 Casio FX-83 {ES or GTplus} or FX-85 {ES or GTplus}
 Sharp Writeview EL-W531
 Hewlett Packard HP10S or HP300S
 Texas TI-30IIS

Section - A

- | | | | |
|-----|--|--|---|
| Q1. | Explain the following terms:
(a) What is Data structure and its classifications.
(b) Differentiate between linear and non-linear data structures.
(c) Explain Time complexity and space complexity in brief.

Consider the following 3D array:
$a_3 = ([\underline{[10, 11, 12]}, [13, 14, 15], [16, 17, 18]],$ $[\underline{[20, 21, 22]}, [23, 24, 25], [26, \underline{27}, 28]],$ $[\underline{[30, 31, 32]}, [33, 34, 35], [36, 37, 38]]])$ (d) What is the location index of the highlighted/underlined number?
(e) Extract the highlighted/underlined value using negative indexing. | [02]
[02]
[02]

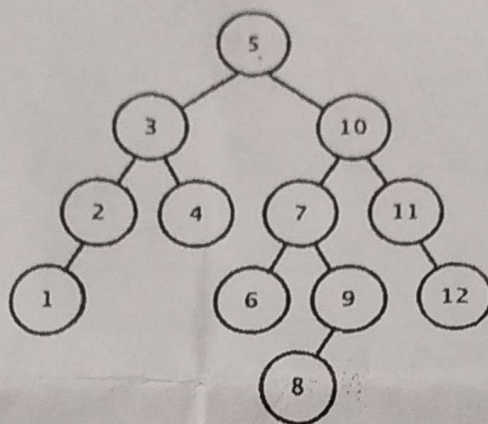
[02]
[02] | [CO1],
[CO2],
[CO4]

[02]
[02] |
|-----|--|--|---|

Section - B

- | | | | |
|-----|---|--------------|---------------------------|
| Q2. | (a) Implement and Explain a doubly linked list.
(b) Explain Quick Sort with the help of an example. | [05]
[05] | [CO1],
[CO2],
[CO3] |
| Q3. | (a) Write a program to insert an entry in a Binary Search Tree.
(b) Write a program to implement queue ADT | [05]
[05] | [CO1],
[CO3],
[CO4] |

Q4.	(a) Explain Stack ADT (b) Write a routine to implement Graphs ADT.	[05] [05]	[CO1], [CO2], [CO3]
Q5.	a) If you wanted to traverse a tree, writing all elements to a file, and later (the next time you ran the program) rebuild the tree by reading and inserting, would an in-order traversal be appropriate? Justify your answer with a suitable example. b) Draw and show the steps of rebalancing an AVL Tree including the balance factor at each node after 5 is removed from the following AVL Tree:	[05] [05]	[CO3], [CO4]



Course Outcomes:

CO1	Learn the c programming language and get in-depth knowledge of Data structures, and their applications in the real world.
CO2	Learn in estimating the complexity of algorithms in terms of space, time, and memory.
CO3	Learn different sorting, searching, and shortest path finding algorithms and also do the complexity analysis of these algorithms
CO4	Gain coding expertise by taking up case studies in data structures and in algorithms.

Question Mapping:

	Questions Mapping	Total Marks	Total Number of Students
CO1	1,2,3,4	40	23
CO2	1,2,4	30	23
CO3	2,3,4	30	23
CO4	1,3,5	30	23